NovaCRM Customer Onboarding Automation: Requirements & Workflow Analysis

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Scope: Part 1 of the Rocketlane FDE take-home. Parts 2–4 (agents, guardrails, tests) are in the repository; this document is the design they implement.

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1. What Priya told us, distilled

NovaCRM (~200 customers) onboards every closed deal by hand. The CS team's process, as described in discovery:

#	Step today	Who	Tool
1	AE emails the CS inbox: customer name + Salesforce opportunity link	AE	Gmail
2	CS creates a project from the standard onboarding template (~15 tasks, 4 phases: Kickoff, Data Migration, Configuration, Go-Live)	CS Manager / CSM	Asana
3	CS creates a shared Slack channel with the customer and schedules the kickoff call	CSM	Slack, calendar
4	Each phase runs off a checklist; tasks are updated as work happens	CSM	Asana
5	Tasks blocked > 3 days should be escalated	CSM → CS Manager	(manual, often missed)
6	At the end: handoff summary, customer moves to the support team	CSM	Docs, support tool

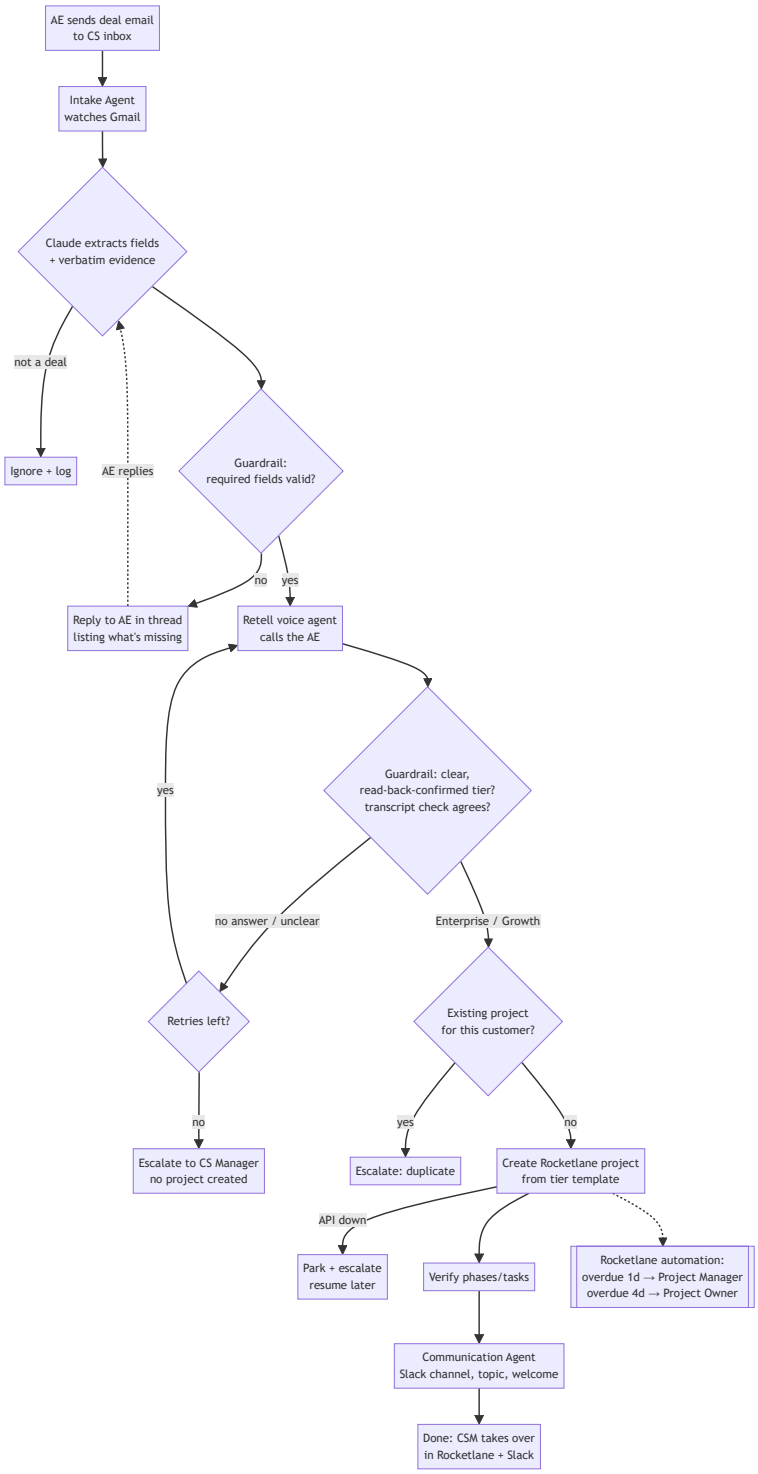
Two plan tiers change the shape of onboarding:

	Enterprise	Growth
Onboarding length	30 days	14 days
CSM model	Dedicated CSM	Pooled CSM

2. Pain points and what causes them

Pain point (Priya's words)	Root cause	What the solution must do
"We sometimes mix up the templates."	Tier is not in the AE's email; a person picks the template from memory.	Make the tier an explicit, verified input. Map tier → template by configuration, never by judgment.
"Escalation falls through the cracks."	Escalation relies on someone noticing a stale task.	Escalate automatically from the system of record (Rocketlane automation), on a fixed schedule.
"Data migration tasks were marked done but the data wasn't verified."	"Done" is self-reported by the CSM; the customer never signs off.	Add an explicit customer data verification sign-off before Go-Live. Built today as a milestone with instructions; a hard dependency and formal customer approval are the production step (§8).
Manual project + channel setup for every deal	Repetitive work across 3 tools	Automate intake → project → channel end to end, with a human only on exceptions.
Inconsistent kickoff experience	Channel names, topics and first messages are ad hoc.	Generate a consistent, tier-specific channel and welcome message.

3. Proposed agentic workflow



3.1 Components

Component	Responsibility	Technology
Intake & Routing Agent	Watch the CS inbox; extract and validate deal details; call the AE to confirm the tier; check for duplicates; create the Rocketlane project from the right template; verify it.	Python, Gmail API, Claude (structured outputs), Retell AI voice agent, Rocketlane REST API
Communication Agent	Create the onboarding Slack channel; set a tier-specific name, topic and purpose; post a personalized welcome; invite the CSM and AE; invite the customer (Slack Connect).	Slack Web API, Claude with a deterministic post-check
Orchestrator	Owns the run state machine, the hand-off between agents, idempotency and resumption.	Python, SQLite

Component	Responsibility	Technology
Rocketlane automation	Overdue-task escalation inside Rocketlane (not an agent).	Rocketlane Automations
Audit log	Every action with timestamp, inputs, outputs, decision and rationale.	JSONL + `novacrm show <run>` timeline

3.2 Why this split

- **The LLM proposes, code decides.** Claude does what it is good at (reading free-form email, judging a transcript, writing a friendly message). Every decision that can cost money or trust (tier, template, CSM, duplicate, retry, escalate) is a deterministic rule in `guardrails.py`, unit-tested.
- **Two agents, one hand-off contract.** The Intake Agent finishes with an `OnboardingHandoff` (deal + confirmed tier + project). The Communication Agent needs nothing else, so it can be re-run on its own (e.g. if Slack fails after the project exists).
- **Rocketlane stays the system of record.** Escalation for overdue work lives in Rocketlane's own automation, so it keeps working even if the agents are offline.

3.3 Run state machine

`RECEIVED → (IGNORED | NEEDS_INFO → SUPERSEDED) → VALIDATED → CALLING(n) → TIER_CONFIRMED | ESCALATED → PROJECT_CREATED | DUPLICATE_ESCALATED | PARKED_RL_DOWN → (CHANNEL_FAILED) → DONE`

Each run is keyed by the Gmail message ID (idempotent), and the Salesforce opportunity link is stored in the project's "Salesforce Opportunity" field, so a crash between "project created" and "state saved" is recovered by adopting the project, not duplicating it. Project creation is never blindly re-posted: a timed-out POST is checked for a project it may have created before any retry (a real bug found and fixed during live testing).

4. Detailed behavior

4.1 Intake: parse and validate

- Inbox: a dedicated CS address (`subburajanperumal+novacrm-cs@gmail.com` in the demo), polled every 15 s; processed mail is labelled (`novacrm-seen`, `-processed`, `-needs-info`, `-escalated`).
- Claude extracts: customer name, contact name, contact email, AE name, Salesforce opportunity URL, close date, notes, each with a **verbatim evidence snippet**.
- **Required:** customer name; customer contact email (valid format, not a NovaCRM or AE address); Salesforce Opportunity URL (Salesforce domain + `006…` ID); sender must be a registered AE with a phone number on file.
- **Grounding check:** any value that does not literally appear in the email is treated as missing. The agent never fills gaps.
- On failure, the agent replies on the AE's thread listing exactly what is missing (Reply-To is the CS inbox). When the AE replies, the whole thread is re-read and the deal continues.

4.2 Intake: confirm the tier by voice

- The email does not carry the tier, and a tier written in the email is ignored for decision purposes: only the AE's spoken confirmation counts.
- A Retell AI voice agent calls the AE. Script: ask "Enterprise or Growth?", read the answer back, require an explicit yes. Never suggest a tier.
- **Transport.** Production: the same agent rings the AE's mobile (number from the AE directory) over a Twilio number, bridged to Retell by SIP, because Retell-owned numbers only dial US numbers. Demo: calling Indian mobiles needs a paid or ID-verified telephony account, so the video uses Retell's web-call

transport: the AE answers an "Incoming call" page and has the same live voice conversation. Decline, or no answer within 45 s, counts as "no answer". The AE's voice is a scripted text-to-speech stand-in in the Enterprise and negative runs, and a real person in the Growth run. Switching to the phone path is one config value (`voice.mode: twilio`), but that path hasn't been run end to end yet.

- Retell's post-call analysis returns `plan_tier` (Enterprise / Growth / Unclear) and `readback_confirmed`. Claude then reads the transcript independently.
- **Accept only if** the call connected, the voice agent captured exactly one tier, the AE confirmed the read-back, and the independent transcript check agrees.
- **No answer / busy / voicemail:** retry up to 3 calls in total (demo: 20 s apart; production: 15 min).
- **Unclear answer:** one re-call (max 2 unclear calls), then escalate.
- **Escalation:** email the CS Manager with the run ID, reason and transcript; label the deal email; post to `#cs-escalations`. No project or channel is created.

4.3 Intake: create the Rocketlane project

	Enterprise	Growth
Template	"NovaCRM Onboarding – Enterprise (30 days)"	"NovaCRM Onboarding – Growth (14 days)"
Due date	start + 29 days	start + 13 days
CSM (team member + "Project Manager" user field)	Dedicated CSM, round-robin from the roster	Pooled CSM queue
Project Owner	CS Manager	CS Manager

- Template IDs come from configuration, keyed by the confirmed tier. The LLM never chooses them.
- Duplicate check first: by Salesforce opportunity ID and by customer name. A match escalates instead of creating.
- After creation the agent lists the project's tasks and compares them with the template (15 tasks, 4 phases). A mismatch is logged and the CS Manager is notified.
- Rocketlane errors: 5xx / 429 / timeouts retry 3 times with exponential backoff; then the run is **parked** and the CS Manager is told; `novacrm resume` finishes it later without calling the AE again. 4xx errors (bad config) escalate immediately.

Template (both tiers, 15 tasks):

Phase	Tasks	Enterprise days	Growth days
Kickoff	AE→CS handoff review · Schedule kickoff call · Kickoff held + success criteria agreed · Customer workspace access provisioned	1–5	1–2
Data Migration	Data export received · Field mapping approved · Test migration run · Production migration run · **Customer data verification sign-off** (milestone)	6–15	3–7
Configuration	Workspace & users configured · Pipelines & custom fields · Integrations connected · Admin training	16–24	8–11
Go-Live	Go-live readiness check (instructions say to wait for the data sign-off) · Go-live + handoff summary to Support	25–30	12–14

Rocketlane adjusts task dates when it imports a template, so individual task dates can differ from these day numbers. The project's start and due dates match the plan.

4.4 Communication Agent

- Channel name: `onb-<customer-slug>-ent|grw` (Slack naming rules; reused if it already exists).
- Topic: `<Tier> onboarding · <N>-day plan · <Dedicated|Pooled> CSM: <name> · Go-live target <date>`.
- Purpose: customer + link to the Rocketlane project.
- Welcome message: drafted by Claude from facts only, then checked: it must name the customer, the right number of days and the CSM, and must not mention the other tier, its timeline, or (for Growth) a dedicated CSM. On any failure the static tier template is used.
- Invites the CSM and AE. Customer invite via Slack Connect is **mocked** in the demo (needs a paid Slack plan; production calls `conversations.inviteShared`).

4.5 Rocketlane automation: overdue escalation

Configured in Rocketlane (Automations), not in code:

- Trigger: task is overdue.
- Block 1: overdue by 1 day → notify the **Project Manager** (the CSM).
- Block 2: overdue by 4 days → notify the **Project Owner** (CS Manager).

The 4-day block has fired for real on a seeded demo task. The 1-day block is configured the same way, but its notification goes to a sample user account, so I haven't seen it arrive.

Priya's original rule, "blocked > 3 days", is about blocked status rather than due dates. I'd add it as a second automation (status = Blocked for 3 days → notify the Project Owner) once NovaCRM confirms how "blocked" is tracked (see Q6).

5. Guardrails (summary)

#	Guardrail	Where
G1	Required fields present, well-formed and **grounded in the email text**; otherwise ask the AE	`guardrails.validate_deal`
G2	Only registered AEs can trigger onboarding (and only they get called)	AE directory
G3	No project until a clear, read-back-confirmed tier, agreed by an independent transcript check; retry, then escalate; never default	`guardrails.decide_tier`
G4	Tier written in the email is ignored for decisions	Intake Agent
G5	Template, duration and CSM come from configuration keyed by tier, never from the LLM	`IntakeAgent.build_payload`
G6	Duplicate check before creation; idempotency per email and per opportunity; no blind re-POST after a timeout	Orchestrator, Rocketlane client
G7	Post-creation verification of the template's tasks	Intake Agent
G8	Welcome message checked for tier consistency, with a safe fallback	`guardrails.check_welcome`
G9	Email and transcript content treated as data, not instructions (prompt-injection resistance)	LLM system prompts
G10	Every action logged with timestamp, input, output, decision and rationale; phone numbers masked	`audit.jsonl`
G11	Secrets only in environment variables; nothing in code or logs	`.env`

6. Assumptions

The deadline left no time for a round-trip with Karthik, so these are the assumptions behind the build. Each is a one-line change if wrong.

1. **Rocketlane replaces Asana** as the project tool (the brief asks for Rocketlane).
2. **AE phone numbers** come from an AE directory (in production: Salesforce User records), because the email doesn't carry them.
3. **Required fields** are customer name, customer contact email and the Salesforce opportunity link; contact name, close date and notes are optional.
4. **The AE is the authority on tier.** A tier written in the email or in Salesforce is not used to decide; in production I would cross-check against the Salesforce opportunity and escalate on a mismatch.
5. **Retry policy:** 3 calls in total, 15 minutes apart in production (20 s in the demo), at most 2 calls that connect without a clear answer.
6. **"Project Manager" = the assigned CSM; "Project Owner" = the CS Manager.**
7. **Dedicated CSMs are assigned round-robin;** Growth deals go to one pooled CSM queue.
8. **Onboarding starts the day the tier is confirmed;** 30/14 days are calendar days, inclusive.
9. **One onboarding project per customer.** An active project for the same customer or opportunity means "escalate, don't create".
10. **Slack:** the channel is created in NovaCRM's workspace and shared with the customer via Slack Connect (mocked here: free plan).
11. **Kickoff scheduling** is announced in the welcome message and owned by the CSM; calendar booking is out of scope for this round.
12. **Handoff summary to support** at the end of onboarding is out of scope for this round (see §8).

7. Questions I'd confirm with NovaCRM

1. Is the AE's email format consistent (template or Salesforce-generated), or free-form?
2. Where do AE phone numbers live today, and may the agent call outside business hours?
3. If the AE stays unreachable, who decides the tier: CS Manager, Sales Ops, or Salesforce data?
4. How many dedicated CSMs are there, and is assignment by capacity, region or round-robin?
5. Should the customer contact be invited to the Rocketlane project at creation, or after kickoff?
6. How is "blocked" tracked today (a status, a tag, a comment)? That decides the blocked-3-days automation.
7. Does the Slack workspace have Slack Connect, and who must approve external channels?
8. Should a Growth customer upgrading to Enterprise mid-onboarding switch templates?

8. What changes in production

Demo choice	Production
Polling Gmail every 15 s	Gmail `users.watch` → Pub/Sub push; a worker queue
Polling Retell for call results	Retell and Twilio webhooks into the orchestrator
Retell web call answered on a local page	Twilio (or local carrier) number ringing the AE's mobile; business-hours calling window, local caller ID
SQLite run store, local JSONL audit	Postgres + a managed log/trace store (OpenTelemetry), retention policy
`.env` secrets	Secrets manager, per-environment keys, rotation
Slack Connect invite mocked	Paid Slack plan, `conversations.inviteShared` + admin approval flow

Demo choice	Production
AE directory YAML	Salesforce API: AE profile, opportunity details, tier cross-check
Manual `resume`	Scheduled retry of parked runs with alerting
Data sign-off is a milestone with instructions	Hard task dependency from the sign-off to Go-Live, plus customer approval in Rocketlane
Out of scope	Handoff-summary agent at Go-Live, kickoff-scheduling agent, blocked-task automation

9. How the deliverables map to the brief

Brief item	Where
Part 1 – requirements	this document
Agent 1 – Intake & Routing	`src/novacrm/agents/intake.py`
Agent 2 – Communication	`src/novacrm/agents/communication.py`
Rocketlane overdue automation	configured in Rocketlane; shown in the video
Part 3 – guardrails	`src/novacrm/guardrails.py`, `escalation.py`, `audit.py`
Part 4 – automated tests	`tests/` (45 tests, offline; `uv run pytest`)
Video – positive and negative runs	Link in the submission email